

CSE-601
B.TECH (CSE) VI SEMESTER
EXAMINATION
Data Science Toolkit

Maximum Marks: 70

Time Allowed: 3 Hours

Note:- Attempt all questions. Internal choice is given.

SECTION – A

I. Short Answer Type Questions (Attempt Any Five) [5×6=30]

- i. What are the main roles in a Data Science project team?
- ii. Define Skewness and Kurtosis in descriptive statistics.
- iii. Write any two differences between NumPy and Pandas.
- iv. What is the function of Logistic Regression in classification problems?
- v. Write short notes on Time Series Regression.
- vi. What are factors and lists in R? Give examples.
- vii. Mention any five features of Power BI and GitHub used in data science.

SECTION – B

II. Long Answer Type Questions (Attempt Any Four) [4×10=40]

- i. Explain the stages in a Data Science project. How is data preprocessing done using cleaning, integration, transformation, and reduction?
- ii. Write a detailed note on **Supervised vs Unsupervised learning** with examples. Explain any two algorithms from each category.
- iii. What is TensorFlow? Explain how **Keras and PyTorch** can be used for **sentiment analysis** or **style transfer**.
- iv. Explain the data structures in R with examples. How can R be used for visualization and classification tasks?
- v. Compare any **three data analytics software tools**: (Weka, Orange, RapidMiner, Minitab, PowerBI). Discuss their advantages and use cases.
- vi. What are **ARIMA models**? Describe the steps of identifying and estimating an ARIMA model for time series forecasting.